Yi's Lemma and a SURE Form for Ledoit–Wolf Shrinkage

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1 Motivation

Under the guidance of Professor Rosenman, the first stage of this project studied shrinkage in the classical normal-means problem. For the linear rule $\hat{\theta} = (1 - \lambda)X$, the correction term is $g(X) = -\lambda X$, and most terms in Stein's unbiased risk estimate (SURE) simplify directly, so the MSE-optimal amount of shrinkage can be displayed graphically across different dimensions and directions. The James–Stein rule $\hat{\theta} = \big(1 - c/\|X\|^2\big)X$ admits the same kind of simplification and can be plotted in the same way.

The central question of the thesis is whether the same program carries over to covariance estimation: can the Ledoit–Wolf (LW) shrinkage estimator be written in SURE form? This note derives the key identity, due to Yi and Zou, and applies it to the LW estimator.

2 Setup

Let $\Sigma = (\sigma_{ij})$ be the true $p \times p$ covariance matrix, $S = (s_{ij})$ the sample covariance matrix, and $\hat{\Sigma} = (\hat{\sigma}_{ij})$ any estimator of Σ. The loss is the squared Frobenius norm. Recall that for a matrix A,

$$\|A\|_F = \Big(\sum_{i,j} |a_{ij}|^2\Big)^{1/2} = \sqrt{\operatorname{tr}(A^H A)} = \Big(\sum_k \sigma_k^2(A)\Big)^{1/2},$$

where A^H is the conjugate transpose and $\sigma_k(A)$ are the singular values of A. Because the loss is the *squared* norm, the square root drops out and the risk decomposes entrywise:

$$R(\hat{\Sigma}) = \mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \sum_{i,j} \mathbb{E}(\hat{\sigma}_{ij} - \sigma_{ij})^2.$$

Assumption. The sample covariance is unbiased, $\mathbb{E}S = \Sigma$, that is, $\mathbb{E}s_{ij} = \sigma_{ij}$ for every (i, j).

Remark. This holds, for example, when the daily observations $x_1, \ldots, x_T \in \mathbb{R}^p$ (one vector per day t) are i.i.d. across days with mean zero and $S = \frac{1}{T}\sum_t x_t x_t^\top$, since then $\mathbb{E}s_{ij} = \frac{1}{T}\sum_t \mathbb{E}[x_{ti}x_{tj}] = \sigma_{ij}$. If the mean is estimated, the demeaned estimator needs the divisor $T - 1$ instead of T to stay unbiased. No distributional assumption (such as normality) is needed.

3 Yi's lemma

Lemma 1 (Yi and Zou)**.** *Under the assumption $\mathbb{E}S = \Sigma$, for any estimator $\hat{\Sigma}$,*

$$\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \mathbb{E}\|\hat{\Sigma} - S\|_F^2 - \sum_{i,j} \operatorname{Var}(s_{ij}) + 2\sum_{i,j} \operatorname{Cov}(\hat{\sigma}_{ij}, s_{ij}).$$

Proof. Fix an entry (i, j) and write $\hat{\sigma}$, s, σ for $\hat{\sigma}_{ij}$, s_{ij}, σ_{ij}. Adding and subtracting s,

$$\hat{\sigma} - \sigma = (\hat{\sigma} - s) + (s - \sigma),$$

so

$$(\hat{\sigma} - \sigma)^2 = (\hat{\sigma} - s)^2 + (s - \sigma)^2 + 2(\hat{\sigma} - s)(s - \sigma).$$

We take expectations term by term.

First term. $\mathbb{E}(\hat{\sigma} - s)^2$ is kept as it is.

Second term. Expanding and using $\mathbb{E}s = \sigma$,

$$\mathbb{E}(s - \sigma)^2 = \mathbb{E}s^2 - 2\sigma\,\mathbb{E}s + \sigma^2 = \mathbb{E}s^2 - \sigma^2 = \mathbb{E}s^2 - (\mathbb{E}s)^2 = \mathrm{Var}(s).$$

Cross term. Distributing,

$$\mathbb{E}\big[(\hat{\sigma} - s)(s - \sigma)\big] = \mathbb{E}\big[\hat{\sigma}(s - \sigma)\big] - \mathbb{E}\big[s(s - \sigma)\big].$$

For any random variables A and B,

$$\mathrm{Cov}(A, B) = \mathbb{E}\big[(A - \mathbb{E}A)(B - \mathbb{E}B)\big] = \mathbb{E}\big[A(B - \mathbb{E}B)\big] - \mathbb{E}A \cdot \mathbb{E}\big[B - \mathbb{E}B\big] = \mathbb{E}\big[A(B - \mathbb{E}B)\big],$$

since $\mathbb{E}[B - \mathbb{E}B] = 0$. Taking $B = s$, so that $\mathbb{E}B = \sigma$, and first $A = \hat{\sigma}$ and then $A = s$, gives

$$\mathbb{E}\big[\hat{\sigma}(s - \sigma)\big] = \mathrm{Cov}(\hat{\sigma}, s), \qquad \mathbb{E}\big[s(s - \sigma)\big] = \mathrm{Cov}(s, s) = \mathrm{Var}(s).$$

Hence the cross term equals $\mathrm{Cov}(\hat{\sigma}, s) - \mathrm{Var}(s)$.

Combining.

$$\mathbb{E}(\hat{\sigma} - \sigma)^2 = \mathbb{E}(\hat{\sigma} - s)^2 + \mathrm{Var}(s) + 2\,\mathrm{Cov}(\hat{\sigma}, s) - 2\,\mathrm{Var}(s) = \mathbb{E}(\hat{\sigma} - s)^2 - \mathrm{Var}(s) + 2\,\mathrm{Cov}(\hat{\sigma}, s).$$

Summing over all (i, j), and noting that $\sum_{i,j} \mathbb{E}(\hat{\sigma}_{ij} - s_{ij})^2 = \mathbb{E}\|\hat{\Sigma} - S\|_F^2$, yields the lemma. □

Remark. The identity holds for *every* estimator $\hat{\Sigma}$, whatever function of the data it is. All three terms are summed over (i, j); the first sum simply recombines into a Frobenius norm. The estimator enters only through $\mathbb{E}\|\hat{\Sigma} - S\|_F^2$ and the covariance term $\sum_{i,j} \mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij})$, which plays the role of the divergence term in Stein's identity.

4 Application to Ledoit–Wolf shrinkage

The LW estimator shrinks S toward a target $F = (f_{ij})$:

$$\hat{\Sigma} = \delta F + (1 - \delta)S, \qquad \text{so} \qquad \hat{\sigma}_{ij} = \delta f_{ij} + (1 - \delta)s_{ij}.$$

We evaluate the three terms of Lemma 1.

First term. Since $\hat{\Sigma} - S = \delta(F - S)$,

$$\mathbb{E}\|\hat{\Sigma} - S\|_F^2 = \delta^2\,\mathbb{E}\|F - S\|_F^2.$$

Covariance term. By bilinearity of covariance,

$$\operatorname{Cov}(\hat{\sigma}_{ij}, s_{ij}) = \delta \operatorname{Cov}(f_{ij}, s_{ij}) + (1-\delta)\operatorname{Var}(s_{ij}).$$

Combining. The variance terms collect as $-\operatorname{Var}(s_{ij}) + 2(1-\delta)\operatorname{Var}(s_{ij}) = (1-2\delta)\operatorname{Var}(s_{ij})$, so the risk as a function of the shrinkage intensity is

$$R(\delta) = \delta^2\, \mathbb{E}\|F - S\|_F^2 + (1-2\delta)\sum_{i,j}\operatorname{Var}(s_{ij}) + 2\delta\sum_{i,j}\operatorname{Cov}(f_{ij}, s_{ij}).$$

5 The SURE form

A SURE is any statistic, computable from the data, whose expectation equals the risk: $\mathbb{E}[\mathrm{SURE}(\delta)] = R(\delta)$. Each term of $R(\delta)$ therefore has to be replaced by an unbiased, computable counterpart. The first term needs no work, since $\|F - S\|_F^2$ is computed directly from S and F and is unbiased for $\mathbb{E}\|F - S\|_F^2$. Given unbiased estimators $\widehat{\operatorname{Var}}(s_{ij})$ and $\widehat{\operatorname{Cov}}(f_{ij}, s_{ij})$,

$$\mathrm{SURE}(\delta) = \delta^2\, \|F - S\|_F^2 + (1-2\delta)\sum_{i,j}\widehat{\operatorname{Var}}(s_{ij}) + 2\delta\sum_{i,j}\widehat{\operatorname{Cov}}(f_{ij}, s_{ij}).$$

6 Open step: estimating the variance and covariance terms

The remaining difficulty is to construct $\widehat{\operatorname{Var}}(s_{ij})$ and $\widehat{\operatorname{Cov}}(f_{ij}, s_{ij})$ in the matrix setting. Everything above rests on two assumptions: the observations have mean zero, and they are i.i.d. across days. Both are essential to the derivation, but neither is fully realistic for financial returns.

Remark (Variance term). With mean-zero i.i.d. data, $s_{ij} = \frac{1}{T}\sum_t y_t$ is an average of the i.i.d. products $y_t = x_{ti}x_{tj}$, so $\operatorname{Var}(s_{ij}) = \frac{1}{T}\operatorname{Var}(y_t)$. The usual unbiased sample variance of the y_t, divided by T, gives an unbiased estimator:

$$\widehat{\operatorname{Var}}(s_{ij}) = \frac{1}{T(T-1)}\sum_{t=1}^{T}\big(x_{ti}x_{tj} - s_{ij}\big)^2.$$

Remark (Covariance term). If the target F is fixed (does not depend on the data), then $\operatorname{Cov}(f_{ij}, s_{ij}) = 0$ and this term vanishes. For data-dependent targets, such as the scaled identity $F = \nu I$ with $\nu = \operatorname{tr}(S)/p$, the term is nonzero and must be estimated; this is the step still to be worked out for general targets.